

automated driving, especially in applications where sufficient braking distance is safety-critical and challenging to achieve with existing technologies. Aeva's unique frequency-modulated continuous wave 4D lidar architecture has superior sensitivity to detect a single photon, without being sensitive to sunlight or other interference.

Aeva Technologies  
www.aeva.ai

## Processor for Edge AI

The M1076 Mythic analogue matrix processor (AMP) delivers up to 25 TOPS in a single chip for high-end edge AI applications. It integrates 76 AMP tiles to store up to 80M weight parameters and execute matrix multiplication operations



without any external memory. These features allow it to deliver the AI compute performance

of a desktop GPU while consuming up to 1/10th of the power—all in a single chip. Each Mythic ACE is complemented by a digital subsystem that includes a 32-bit RISC-V nano processor, SIMD vector engine, 64kB SRAM, and a high-throughput network-on-chip router. Mythic AMP is designed to provide powerful compute in small edge devices and easily scale to deliver server-class compute in data centres.

Mythic  
www.mythic-ai.com

## Surround view cameras

Thanks to Magna's new surround view cameras and electronic control units, global automakers will soon be able to offer 3D surround-view systems—a



driver-assistance technology found mainly in luxury-class vehicles—to more vehicles. Magna's multi-camera system provides a high-resolution, 360-degree

field of view around the vehicle. The domain controller then creates a 3D surround-view image by processing the four camera images, rendering a seamlessly stitched 3D view of the vehicle surroundings. The system helps drivers park in even the tightest of spaces and provides an enhanced level of comfort and convenience. It also delivers data to improve the performance of other vehicle systems such as emergency braking and automatically detects and warns the driver when the camera lens is impeded by snow, ice, dirt or raindrops.

Magna  
www.magna.com

## HD thermal imaging camera

This advanced thermal imaging camera with an infrared sensor has a high distinctive sensitivity to perceive farther



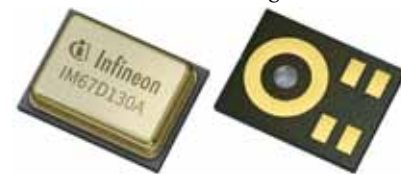
and subtler heat sources. The thermal sensor captures heat from the object and converts it into a signal to form a thermal image. Due to the unevenness of the image, the integrated algorithm enhances the original thermal image to realise the visibility and feasibility of object recognition. Besides, compensation is made for the blind pixels that appear in the imaging process. Therefore, the final thermal image shows more contrast and completeness. The thermal imaging camera has the important feature of ultra-low power consumption, which helps extend the device's battery life. The 384 × 288 infrared detectors consume only 70mW under the working conditions of 60Hz digital output as compared to 220mW of a similar product.

InfiRay  
www.infiray.com

## Low-noise MEMS microphone

The XENSIV IM67D130A is a new device that comprises high-end, low-noise MEMS microphone technology to

address the need for automotive applications. The device has an increased operating temperature range from -40°C to +105°C to enable various use cases in harsh automotive environments. The high acoustic overload point of 130dB SPL allows the microphone to capture distortion-free audio signals in loud



environments. Because of that, it can be placed inside or outside a vehicle. IM67D130A is ideal for in-cabin applications such as hands-free systems, emergency calls, in-cabin communication, and active noise cancellation. It is also suited for exterior applications such as siren or road condition detection. These features enable the use of sound as a complementary sensor for advanced driver assistance systems and predictive maintenance.

Infineon Technologies  
www.infineon.com

## All-in-one navigation system

The advanced inertial navigation system includes the 3DMGQ7-GNSS/INS, which provides a centimetre-level position accuracy when accompanied by the company's 3DMRTK modem and SensorCloud RTK corrections service. It features dual multiband GNSS receivers, low-noise and low-drift MEMS inertial sensors,



and a robust adaptive Kalman filter. The 3DMRTK provides the simplicity of

a cellular connection to Parker LORD's SensorCloud RTK base station network, replacing traditional radio based stations. By removing the need for base station infrastructure, it minimises an integrator's required time to market. The SensorCloud RTK provides seamless cloud based network RTK corrections without the need for customer-supplied base stations and includes a cellular data plan with network RTK coverage.

Parker LORD MicroStrain Sensing  
www.microstrain.com